

Soft Short-Circuit Fault Diagnosis for Vehicular Battery Packs With Interpretable Full-Dimensional Statistical Analytics

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Abstract—Severe thermal runaway is induced in the majority by short-circuit faults inside battery packs in vehicular applications. In this article, the challenging issue of interpretable soft short-circuit fault diagnosis for lithium-ion battery packs is investigated comprehensively. By using the streaming differential matrices from improved virtual voltage measurement, a full-dimensional statistical analysis scheme considering the random occurrence of a soft short-circuit is proposed to characterize the fault signatures. On this basis, an eigenvalue-decomposition-based fault detection indicator is designed, and the quantitative threshold correlation between the designed fault indicator and the original fault signature is derived such that the interpretable fault detection control limit self-commissioning can be achieved. Furthermore, the detected fault can be localized by evaluating the eigenvalue contributions to the designed fault detection indicator. Finally, a test platform with commercial lithium-ion phosphate cells in a compact battery pack is constructed to validate the approach. Experiments conducted with the World Light Vehicle Test Cycle confirm that millivolt-level short-circuit fault diagnosis can be achieved within minutes, and the fault detection and localization procedures are complementary such that the false alarm rate is minimized. It is robust against both the momentary and cumulative short-circuit conditions even with a maximum 2-C current within almost 83% state-of-charge range.

Index Terms—Electric vehicle (EV), fault diagnosis, lithium-ion battery (LiB), soft short-circuit (SSC), statistical analysis.

I. INTRODUCTION

IN THE face of escalating environmental pollution and energy crisis, electric vehicles (EVs) emerge as a viable solution for achieving a future carbon-neutral society. Lithium-ion batteries

(LiBs), with their high power density and extended service life, have become the cornerstone for EV deployment [1]. However, the safety of LiBs has gained significant attention due to occasional accident occurrences resulting from LiBs failures [2].

Thermal runaway in LiBs is pinpointed as the primary cause behind the majority of fires and explosions in EVs, necessitating urgent measures for enhanced safety. This phenomenon can be triggered by various forms of abuse, including electrical, mechanical, and thermal abuses [3], all of which may lead to a short circuit (SC) in LiBs [4]. It is important to note that at the onset of an SC fault, the symptoms might be subtle, typically characterized as a soft SC (SSC). To prevent more catastrophic outcomes, incorporating effective SSC fault diagnosis strategies within the battery management system (BMS) is imperative [5].

A. Literature Review

The SSCs can be primarily divided into momentary SSC and cumulative SSC, rather than internal and external SSCs due to their similar fault characteristics, such as voltage drop and temperature rise [6]. In recent years, numerous methods have been proposed for the SSC fault diagnosis of LiBs, broadly falling into two categories: 1) model-driven and 2) data-driven methods.

Regarding the model-driven SSC fault diagnosis, it is generally transformed into model parameter estimation, as the abnormal voltage signature due to extra charge depletion can be captured by model-driven residual evaluation with standard one-to-one correspondence voltage sensing [7]. Within this framework, high-precision estimation models are developed for cell-level condition monitoring, drawing on principles of equivalent circuit [8], electrochemical [9], and electrothermal coupling [10]. To identify incipient SC faults, a method based on the state-of-charge (SOC) estimation of the faulty cell is proposed in [11], where the incipient SC resistance is identified to indicate the fault severity. With iterated SC resistance estimation by the designed adaptive neural observer, a delicate diagnosis is investigated in [12] to detect SSC fault occurrences in the hour scale. Apart from SC resistance estimation, the SC current during constant voltage charging is estimated in [13] for SSC diagnosis. In [14], a capacity-estimation-based SSC fault diagnosis method is proposed, facilitating offline quantitative

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evaluation to distinguish SC fault cells from those with diminished capacity over one discharge–charge cycle.

In these samplewise model-driven methods, fault detection and localization functions are intertwined by concurrently estimating the SSC fault parameters across all series-connected cells [15]. The fault parameter estimation accuracy is maintained under experimental conditions without the necessity to differentiate between the momentary and cumulative faults. Nevertheless, these methods rely on effective SOC variation evaluation based on detailed cell models. The reliability of SC resistance and cell capacity estimations hinges on sufficiently large SOC or voltage changes before initiating the iterative estimation process, and the timeliness is greatly reduced in turn. Notably, more effort is still required for online model-driven SSC fault identification, especially when comprehensively accounting for inherent parameter uncertainties [16] and cell inconsistencies [11], which can result in significant fault parameter estimation residuals.

In the realm of data-driven SSC fault diagnosis, methods that primarily utilize voltage, current, and temperature signals with different measurement schemes have also been widely studied for LiB packs, thus largely getting rid of the reliance on detailed battery models. With the standard voltage measurement topology, a quantitative SSC fault diagnosis method is investigated in [17] to identify the equivalent resistance based on capacity loss evaluation by determining the voltage plateaus. Nonetheless, data-driven capacity estimation is still challenging for vehicular applications due to frequent transitions between charging and discharging conditions [14]. Recently, benefiting from reduced computational complexity from windowed streaming data analytics, efficient online diagnosis for SSC fault with statistics-based data-driven methodologies has been widely carried out by the academic community. Based on the kernel principal component analysis (PCA) for standardized measurement of cell-voltage differences, the T^2 and Q test statistics are further tailored in [18] for SSC diagnosis of LiB packs. The SSC sensitivity of this method is dependent on reasonable data dimensionality-reduction [19]. Beyond PCA-based statistical analysis incorporating feature-space partitioning, a variety of full-dimensional statistics-based methods have been developed for SSC diagnosis of LiB packs. A trilayered diagnosis method based on statistical analysis is investigated in [20] to detect and localize the momentary and cumulative SSC faults with an improved K -means algorithm. By using extremum sample entropy, an applicable SSC fault diagnosis method is proposed in [21] with standard voltage measurement. With the same measurement paradigm, a diagnosis strategy predicated on voltage cosine similarity has been developed in [22]. By using the improved correction coefficient (CC) method, online diagnosis for SC fault is investigated in [23] with an interleaved voltage measurement scheme. On this basis, a voltage-correlation-based method is further proposed in [24] incorporating the PCA technique. These methods have been confirmed to be sensitive to momentary SSC faults.

With advancements in artificial intelligence, machine-learning-based data-driven methods have increasingly become prominent in fault diagnosis of vehicular batteries. Unsupervised training is integrated into the identification of mixed kernel

parameters in [25] for SSC fault diagnosis based on nonlinear cross-cell monitoring. A local-outlier-factor-based machine learning method is proposed in [26] to diagnose the SSC fault for LiB packs, in which the incremental capacity analysis is introduced to extract features from massive standard measurement voltage-capacity datasets. Additionally, ensemble learning has been employed in [27] to facilitate fault information fusion across various window widths with standard voltage measurement, such that more comprehensive and robust diagnosis results can be obtained after natural SSC fault isolation.

In these data-driven fault diagnosis methods, the majority rely on standard voltage measurement, compatible with commercial BMS. To achieve the same hardware nonredundancy measurement, interleaved voltage measurement can be applied to naturally differentiate SSC faults from other electrical faults in a multifault diagnosis context [28]. Among them, fault detection and localization typically constitute two interdependent and sequential processes, where the fault detection stage determines the presence of an SSC fault, and it dominates the course of SSC fault diagnosis. This noncomplementary configuration has to degrade the SSC fault diagnosis performance with additional significant computation overhead for subordinate fault localization. Thanks to insightful feature extraction for data-driven SSC fault diagnosis from effective data mining and statistic analytics with appropriate measurement schemes, comparative performance with model-driven SSC fault diagnosis methods can be achieved. Nevertheless, the diagnosis performance of windowwise data-driven methods significantly hinges on prior knowledge due to the inherent uninterpretable characteristics of data-driven modeling. The availability of datasets under clearly defined normal and abnormal conditions is essential for fault signature analysis, and misdiagnosis is inevitable with the statistical indicators designed in the original measurement space, especially in the case of cumulative SSC fault.

B. Motivations and Contributions

To date, numerous fault diagnosis methods have been developed to address the detection and localization of SSC faults. Among them, fault detection and localization are coupled with each other for model-driven methods while cascaded fault detection and localization is the general case in data-driven methods.

Nevertheless, few works have focused on the fundamental fault detectability issue aiming at interpretable fault diagnosis for vehicular LiB packs, which characterizes the quantitative threshold correlation between the designed fault indicator and the original fault signature. This challenging problem is more critical for data-driven methods without fault parameter estimation. The fault-signature threshold is the tolerated fault amplitude with physical significance while the fault indicator threshold is the control limit with mathematical significance to trigger the fault alarm. Indeed, the SSC fault signature can be easily masked by inherent parameter-inconsistency-induced fluctuations imposed on measurable signals and their statistics, and the SSC fault diagnosis performance has to be inevitably degraded in turn. Whether the standard measurement scheme in commercial

BMS can be improved to enhance the SSC fault diagnosis performance has not gained enough attention. In the existing studies, to tolerate the inherent parameter-inconsistency-induced measurement fluctuation uncertainties, posterior experience-based user-defined configuration determines the fault estimation triggering in model-driven methods. Similarly, posterior experience from normal distribution hypothesis or further kernel density estimation (KDE) for limited fault signature streaming dominates the fault indicator design and control limit derivation for data-driven methods. Also, the quantitative correlation between fault signature and detection indicator/triggering in these methods remains ambiguous. Crucially, this quantitative correlation is directly related to the reasonable control limit self-commissioning which dominates the SSC fault sensitivity and diagnosis robustness.

By investigating the pivotal issue of fault detectability in data-driven SSC fault diagnosis for vehicular LiB pack applications with standard measurement voltage streaming, a comprehensive fault diagnosis method based on interpretable full-dimensional statistical analytics is proposed in this article. The main contributions are outlined as follows.

- 1) An improved virtual voltage measurement scheme is investigated to minimize parameter-inconsistency-induced differential voltage fluctuations by combining with efficient power electronics-based near-cut-off voltage-aligned (NCVA) operation.
- 2) Complementary fault detection indicators and fault localization are well-designed based on high-order statistics, and the control limit self-commissioning can be achieved with interpretable fault detectability in the statistical sense.
- 3) A universal SSC fault simulation method for both momentary and cumulative SSC fault conditions is investigated to validate the superior fault diagnosis performance of the proposed method.

The rest of this article is organized as follows. Section II delves into the improved virtual voltage measurement scheme and the SSC fault detection indicator design. Section III elucidates the comprehensive SSC fault diagnosis scheme. Section IV illustrates the experiment setup and investigates the fault injection method. The validation is demonstrated in Section V. Finally, Section VI concludes this article.

II. FAULT DETECTION SCHEME DESIGN

In this section, the mechanism analysis regarding differential voltage fluctuations involved in the standard voltage measurement topology in commercial LiB packs is presented first. Furthermore, an improved virtual voltage measurement scheme is introduced to minimize the fluctuation spikes of differential

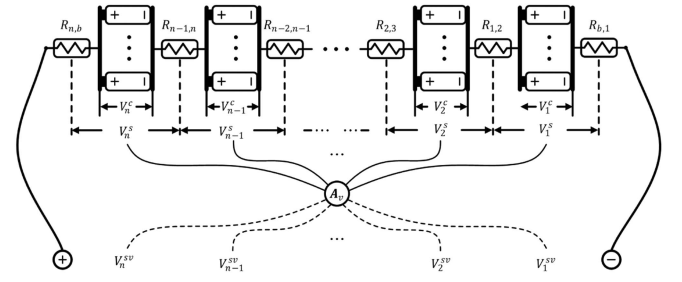


Fig. 1. Schematic of the virtual voltage measurement scheme.

voltage streaming. On this basis, an eigenvalue-decomposition-based fault detection indicator is designed to capture the SSC fault signature with the rigorous theoretical derivation of its fault detectability.

A. Differential Voltage Fluctuation Mechanism Analysis

In the standard voltage measurement topology as illustrated in the upper portion of Fig. 1, the cell voltage sensors are cross-cell connected. For the battery pack with n series-connected cells, the i th sensor is linked to both the positive and negative electrodes of cell i where $i \in [1, n]$. Consequently, the voltage sampling expression at time index k is represented as (1) shown at the bottom of this page, where $\{V_{i,k}^s\}$ denotes the sampled voltage, $\{V_{i,k}^c\}$ corresponds to the cell voltages and $I_{p,k}$ signifies the pack current.

As outlined in (1), for $i \in (2, n-1)$, the voltage drop across the connection resistor $R_{i-1,i}$ and $R_{i,i+1}$ is incorporated into the voltage sample $V_{i,k}^s$. Meanwhile, the voltage drops across the busbar connection resistors $R_{n,b}$ and $R_{b,1}$ are factored into the voltage samples $V_{1,k}^s$ and $V_{n,k}^s$, respectively. It is important to note that under normal operating conditions, the connection resistors between individual cells are generally constant and nearly identical for commercially compact LiB packs. In contrast, the cell voltage is time-varying with SOC and duty-cycle current as

$$V_{i,k}^c = V_{oc,i,k}(\text{SOC}_i) - I_{p,k}R_{s,i,k}(\text{SOC}_i) \quad (2)$$

where $V_{oc,i,k}$ and $R_{s,i,k}$ denote the open-circuit voltage (OCV) and internal resistance (IR) of i th LiB cell, and SOC_i is the corresponding SOC.

To reduce the duty-cycle-related voltage fluctuations during the normal operating condition, the differentiating operation with the sampling voltage $\{V_{i,k}^s\}$ is straightforward, which can

$$\begin{bmatrix} V_{1,k}^s \\ V_{2,k}^s \\ \vdots \\ V_{n-1,k}^s \\ V_{n,k}^s \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix} \begin{bmatrix} V_{1,k}^c \\ V_{2,k}^c \\ \vdots \\ V_{n-1,k}^c \\ V_{n,k}^c \end{bmatrix} - \frac{1}{2}I_{p,k} \begin{bmatrix} 1 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & 0 & \cdots & 1 & 1 \end{bmatrix} \begin{bmatrix} R_{b,1} \\ R_{1,2} \\ R_{2,3} \\ \vdots \\ R_{n-1,n} \\ R_{n,b} \end{bmatrix}. \quad (1)$$

be formulated and simplified as

$$\Delta \mathbf{V}_k^s \approx \mathbf{A}_d \mathbf{V}_k^c = \mathbf{A}_d (\mathbf{V}_{oc,k} - I_{p,k} \mathbf{R}_{s,k}) \quad (3)$$

where $\Delta \mathbf{V}_k^s$ is the differential voltage vector, $\mathbf{V}_{oc,k} = [V_{oc,1,k}, V_{oc,2,k}, \dots, V_{oc,n,k}]^T$, and $\mathbf{R}_{s,k} = [R_{s,1,k}, R_{s,2,k}, \dots, R_{s,n,k}]^T$. $\mathbf{A}_d$ is the differential operation matrix, and the standardized adjacent version can be expressed as

$$\mathbf{A}_d = \begin{bmatrix} 1 & -1 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -1 \\ -1 & 0 & \cdots & 0 & 1 \end{bmatrix}. \quad (4)$$

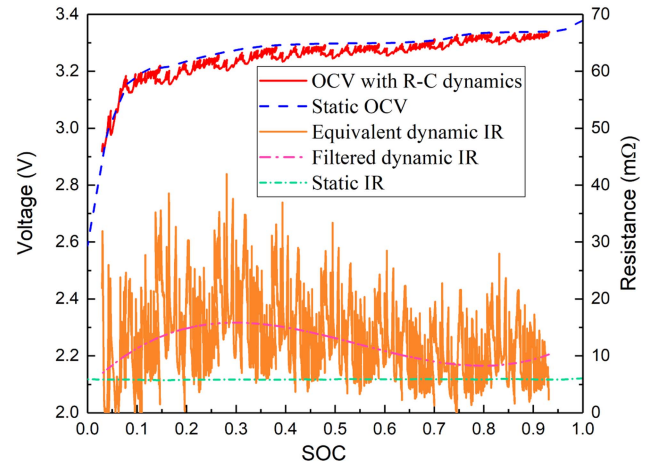
However, the inherent cell parameter inconsistencies involved in $\mathbf{V}_{oc,k}$ and $\mathbf{R}_{s,k}$ are difficult to eliminate, although effective sorting strategies are applied before battery grouping. Considering the typical IR and OCV curves of lithium-ion phosphate cells as depicted in Fig. 2(a), the static IRs vary slightly, and their consistency can be flexibly guaranteed by battery grouping. Nevertheless, by eliminating the impact of static OCV, the equivalent dynamic IR resulting from $R-C$ dynamics is SOC dependent and its filtered curve changes intensively. Thus, satisfactory dynamic IR consistency among individual cells is difficult to maintain.

In the meanwhile, the OCV exhibits a flat voltage plateau [17], yet sharply declines once the SOC falls below 8%. Given inherent capacity inconsistency among cells, there is noticeable SOC bias, leading to the LiB pack being considered fully charged or discharged in actual operation once any cell reaches its cutoff voltage [29]. Note that, for an equivalent $R-C$ network with a small resistor connected in parallel with a large capacitor, the resistive current flowing through the small resistor corresponding to the above filtered dynamic IR generally varies not significantly. Then, we postulated two shifted filtered dynamic IRs with a similar curve shape as the filtered dynamic IR in Fig. 2(a). Consequently, it can result in significant differential voltage fluctuation ΔV_{IR} across the SOC range where SOC_{LL} and SOC_{HL} are denoted as SOC boundary values corresponding to the significant change of OCV, and its filtered curve is depicted in Fig. 2(b). It becomes evident that an additional voltage residual ΔV_{p-p} is superimposed on the differential voltage streaming if the two cell voltages are not initially “aligned” when considering dynamic IR and capacity inconsistencies during the cyclic operation. Also, these differential voltage spikes, varying with the duty-cycle current, are significantly pronounced during high current-ratio (CR) operations.

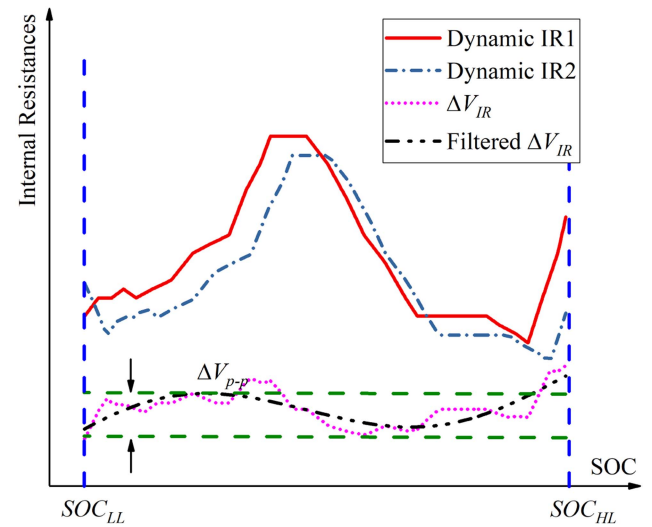
It can be confirmed that the SSC fault diagnosis performance based on the conventional standard voltage measurement scheme has to be degraded, as the SSC fault signatures can be easily masked by these parameter-inconsistency-induced differential voltage fluctuations.

B. Improved Virtual Voltage Measurement Scheme

To more effectively identify the SSC fault signature, a virtual voltage measurement scheme is proposed in this section,



(a)



(b)

Fig. 2. Impact illustration of inherent parameter inconsistencies. (a) Typical OCV and estimated IR curves. (b) Differential voltage fluctuations superimposed by dynamic IR and capacity inconsistencies.

which is improved from both temporal and spatial dimensions, respectively.

From the temporal perspective, a modified NCVA operation with minimum cell-balancing performance overhead is explored to reduce the parameter inconsistency-induced differential voltage fluctuations. That is, once the cell voltages are balanced (aligned) around the near-cut-off voltage, the cell balancing can be disabled for the remainder of the discharging cycle and the ensuing charging–discharging loop. As the passive balancing methods have to remove excess energy of the higher voltage cell using heat dissipation on the parallel-switching resistors or MOSFETs, a more efficient power-electronics-based active balancing configuration is recommended for this NCVA operation. Through extensive testing on a compact experimental lithium-ion phosphate battery pack, the aligned near-cut-off voltage is roughly selected as 2.94 V, which aligns with nearly half of the SOC around the lower OCV turning point in Fig. 2(a).

Note that, the OCV divergence tends to be significant at low and high SOC levels [30]. As rapid voltage divergence in $[0, \text{SOC}_{LL}]$ and $[\text{SOC}_{HL}, 1]$ is inevitable although active balancing is configured and activated, the introduced NCVA operation does not adversely affect the amount of available energy drawing from the LiB packs.

Furthermore, to reduce the inconsistency effect from spatial dimension, additional virtual transformation needs to be applied for (1) considering that the highest OCV and IR correlations among individual cells are probably nonadjacent, as demonstrated in the bottom half of Fig. 1. The voltage sampling vector in (1) can be rearranged as

$$\mathbf{V}^{\text{sv}} = \mathbf{A}_v \mathbf{V}^s \quad (5)$$

where $\mathbf{V}^s = [V_1^s, V_2^s, \dots, V_n^s]^T$, $\mathbf{V}^{\text{sv}}$ is the virtual voltage measurement vector, and $\mathbf{A}_v$ is the virtual transformation matrix. For the entries $\{a_{i,j}^v\}$ in $\mathbf{A}_v$, they satisfy that

$$a_{i,j}^v = \begin{cases} 1, & V_i^{\text{sv}} = V_j^s \\ 0, & V_i^{\text{sv}} \neq V_j^s \end{cases} \quad i, j \in [1, n] \quad (6)$$

Combined with (3) and (5), the transformed streaming differential voltage vector is formulated as

$$\Delta \mathbf{V}_k^{\text{sv}} = \mathbf{A}_d \mathbf{A}_v \mathbf{V}_k^c \quad (7)$$

where $\Delta \mathbf{V}_k^{\text{sv}} = [\Delta V_{1,2,k}^{\text{sv}}, \Delta V_{2,3,k}^{\text{sv}}, \dots, \Delta V_{n-1,n,k}^{\text{sv}}, \Delta V_{n,1,k}^{\text{sv}}]^T$.

Based on the improved virtual voltage measurement scheme, it is expected that differential voltage fluctuations involved in the standard measurement voltage streaming can be greatly reduced to highlight the SSC fault signature under normal operation conditions.

C. Eigenvalue-Decomposition-Based Fault Detection

Note that, the rank of differential operation matrix $\mathbf{A}_d$ is $n - 1$. Therefore, (7) needs to be split into two differential subvectors $\Delta \mathbf{V}_{1,k}^{\text{sv}}$ and $\Delta \mathbf{V}_{2,k}^{\text{sv}}$ with $n - 2$ overlapped entries. Correspondingly, $\mathbf{A}_d$ needs to be split into two submatrices $\mathbf{A}_{d,1}$ and $\mathbf{A}_{d,2}$.

Then, the fault detection indicator can be designed based on eigenvalue analysis of the covariance matrix for the streaming transformed differential voltage matrices.

Without loss of generality, the individual transformed differential voltage vector can be formulated as

$$\mathbf{x} = \mathbf{A} \mathbf{s} \quad (8)$$

where $\mathbf{x} = \Delta \mathbf{V}_{i,k}^{\text{sv}} \in \mathbb{R}^{(n-1) \times 1}$, $\mathbf{A} = \mathbf{A}_{d,i} \mathbf{A}_v \in \mathbb{R}^{(n-1) \times n}$ is the virtual measurement coefficient matrix, $i \in \{1, 2\}$; $\mathbf{s} \in \mathbb{R}^{n \times 1}$ stands for n -cell voltage sources, and each sample can be considered as independent identically distributed.

For the general multivariate process represented by (8), considering the streaming virtual measurement matrix $\mathbf{X} \in \mathbb{R}^{w \times (n-1)}$ at time index k with sliding window width w

$$\mathbf{X}_k = \begin{bmatrix} x_{k-w+1,1} & x_{k-w+1,2} & \cdots & x_{k-w+1,n-1} \\ x_{k-w+2,1} & x_{k-w+2,2} & \cdots & x_{k-w+2,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ x_{k,1} & x_{k,2} & \cdots & x_{k,n-1} \end{bmatrix} \quad (9)$$

we have

$$\mathbf{X}_k = \mathbf{S}_{1,k} \mathbf{A}^T \quad (10)$$

where $\mathbf{S}_k \in \mathbb{R}^{w \times n}$.

Then, the original virtual measurement matrix can be normalized as

$$\bar{\mathbf{X}}_k = (\mathbf{X}_k - \boldsymbol{\mu}_0^T) \boldsymbol{\Sigma}_0^{-1} = \mathbf{S}_k \bar{\mathbf{A}}^T - \boldsymbol{\mu}_0^T \boldsymbol{\Sigma}_0^{-1} = \bar{\mathbf{S}}_k \bar{\mathbf{A}}^T \quad (11)$$

where $\boldsymbol{\mu}_0 \in \mathbb{R}^{n-1}$ denotes the reference sample mean vector and $\boldsymbol{\Sigma}_0 = \text{diag}\{\sigma_{x_1}, \dots, \sigma_{x_{n-1}}\} \in \mathbb{R}^{(n-1) \times (n-1)}$ denotes the reference sample standard deviation (SD) diagonal-matrix, which can be obtained using sufficient historical datasets under the normal condition. To normalize the voltage source terms, $\boldsymbol{\mu}_s$ denotes the reference source mean vectors. Then, $\boldsymbol{\mu}_0 = \mathbf{A} \boldsymbol{\mu}_s$, $\bar{\mathbf{A}} = \boldsymbol{\Sigma}_0^{-1} \mathbf{A}$, and $\bar{\mathbf{S}}_k = \mathbf{S}_k - \boldsymbol{\mu}_s^T$.

The covariance matrix of $\bar{\mathbf{X}}_k$ can be formulated as

$$\mathbf{C}_k = \mathbf{V}_k \boldsymbol{\Lambda}_k \mathbf{V}_k^T \approx \frac{1}{w} \bar{\mathbf{X}}_k^T \bar{\mathbf{X}}_k \quad (12)$$

where $\boldsymbol{\Lambda}_k = \text{diag}\{\lambda_{1,k}, \lambda_{2,k}, \dots, \lambda_{n-1,k}\}$ and $\mathbf{V}_k$, respectively, denote the eigenvalue diagonal-matrix with descending order and corresponding eigenvector matrix.

Due to the high-order statistical characteristics of abnormal conditions being orthogonal to that of normal ones, it can be characterized by the eigenvalue without a normal component. In this viewpoint, the fault detection performance can be improved with eigenvalue-based full-dimensional analysis for the streaming virtual measurement matrices $\mathbf{X}_k$. Then, the fault detection indicator DI_k is designed with the infinite norm of the normalized eigenvalue $\{\lambda_{j,k}\}$ of $\mathbf{C}_k$ as

$$\text{DI}_k = \max_{1 \leq j \leq n-1} \left| \frac{\lambda_{j,k} - \mu_{\lambda_j^*}}{\sigma_{\lambda_j^*}} \right| \quad (13)$$

where $\mu_{\lambda_j^*}$ and $\sigma_{\lambda_j^*}$ denote the mean and SD of $\{\lambda_j^*\}$, respectively. In this article, the variables with “*” marked are obtained with the fault-free dataset under normal conditions.

Note that, the eigenvalue decomposition implies a rigid rotation within the sliding window such that the transformed components $\bar{\mathbf{X}}_k \mathbf{V}_k$ in the new coordinate system are orthogonal and have maximum sample variance subject to corresponding spatial-temporal constraints. By using this simple full-dimensional linear transformation, the statistical information among measurements is fully utilized with the proposed data-driven-based fault detection indicator, apart from the statistical information under normal conditions, and the real-time space partition can be effectively avoided.

Indeed, performing the individual eigenvalue decomposition in the formulated (13) in each sliding window is computationally intensive due to the multiple matrix operations involved, especially for resource-limited embedded BMS applications. However, it is noteworthy that most entries of adjacent streaming virtual measurement matrices remain unchanged. This “diagonal-plus-rank-one” characteristic [31] facilitates a more efficient algorithm through recursive eigenvalue updating, significantly reducing computational complexity, which was elaborated upon

in our prior work [32]. By applying this rank-one modification-based algorithm, the root finding for involved critical secular functions can be independently iterated in parallel with a quadratic rate of convergence, making the computational cost of the eigenvalues very low [33].

D. Fault Detectability of the Designed Indicator

Here, additive SSC fault is considered, and the fault only occurs at one LiB cell for incipient fault conditions. For the SSC fault, in the presence of fault signature f_{sc} occurs at the l th channel s_l , the differential measurement vector can be described as

$$\mathbf{x} = \mathbf{A}(\mathbf{s} + \xi f_{sc}) \quad (14)$$

where $\xi \in \mathbb{R}^n$ representing the fault weight, which is nonzero only at the faulty channel with $\|\xi\|_1 = 1$.

Denote $\bar{\mathbf{Y}}_k = \bar{\mathbf{S}}_k \bar{\mathbf{A}}^T$ at time index k . Then, the streaming matrix $\bar{\mathbf{X}}_k$ in sliding window w can be normalized as

$$\bar{\mathbf{X}}_k = (\mathbf{X}_k^* + f_{sc,k} \xi_k^T \mathbf{A}^T) \Sigma_0^{-1} = \bar{\mathbf{Y}}_k + f_k \xi_k^T \bar{\mathbf{A}}^T \quad (15)$$

where $f_{sc,k} \in \mathbb{R}^{w \times 1}$ is the fault vector, which characterizes the fault duration in the sliding window w .

Then, the streaming covariance matrix of $\bar{\mathbf{X}}_k$ can be represented by $\mathbf{C}_k = \sum_{i=1}^3 \mathbf{C}_{i,k}$, where

$$\begin{cases} \mathbf{C}_{1,k} = w^{-1} \bar{\mathbf{Y}}_k^T \bar{\mathbf{Y}}_k \\ \mathbf{C}_{2,k} = w^{-1} \bar{\mathbf{Y}}_k^T f_{sc,k} \xi_k^T \bar{\mathbf{A}}^T + w^{-1} \bar{\mathbf{A}} \xi_{sc,k} f_k^T \bar{\mathbf{Y}}_k \\ \mathbf{C}_{3,k} = w^{-1} \bar{\mathbf{A}} \xi_k^T f_{sc,k}^T f_{sc,k} \xi_k^T \bar{\mathbf{A}}^T. \end{cases} \quad (16)$$

The proposed fault detection indicator, as formulated in (13), is directly related to the eigenvalue distribution of the streaming covariance matrix $\mathbf{C}_k$. Hence, the eigenvalues' expectations need to be investigated to gain insight into the quantitative threshold correlation between the designed fault indicator and the original fault signature.

Although the parameter-inconsistency-induced differential voltage fluctuations can be significantly reduced by the improved virtual voltage measurement scheme, the resulting differential voltage SDs are still generally different from each other. As $\mathbb{E}\{D_k\} \leq \delta_{sc}$ need to be guaranteed for the global fault detectability, we have

$$\mathbb{E}\{\lambda_i\} \leq \mu_{\lambda_i}^* + \delta_{sc} \sigma_{\lambda_i}^* \quad i \in \{1, \dots, n-1\} \quad (17)$$

where δ_{sc} is the control limit for SSC fault detection. According to the additivity of matrix trace, δ_{sc} can be bounded by

$$\begin{aligned} \delta_{sc} &\geq \frac{\sum_{i=1}^{n-1} \mathbb{E}\{\lambda_{i,k}\} - \sum_{i=1}^{n-1} \mu_{\lambda_i}^*}{\sum_{i=1}^{n-1} \sigma_{\lambda_i}^*} \\ &= \frac{\sum_{j=1}^3 \mathbb{E}\{\text{Tr}(\mathbf{C}_{j,k})\} - \sum_{i=1}^{n-1} \mu_{\lambda_i}^*}{\sum_{i=1}^{n-1} \sigma_{\lambda_i}^*}. \end{aligned} \quad (18)$$

Note that $\mathbf{C}_{1,k}$ is the normalized covariance matrix under normal conditions. Due to the eigenvalue expectation satisfying

$\mathbb{E}\{\lambda_{i,k}\} = \mu_{\lambda_{i,k}}^* (i = 1, \dots, n-1)$, we have

$$\mathbb{E}\{\text{Tr}(\mathbf{C}_{1,k})\} = \sum_{i=1}^{n-1} \mu_{\lambda_i}^*. \quad (19)$$

Besides, according to the commutativity between matrix trace and expectation, we have $\mathbb{E}\{\text{Tr}(\mathbf{C}_{2,k})\} = 0$ as $\mathbb{E}\{\bar{\mathbf{Y}}_k^T f_k\} = 0$ for $\mathbf{C}_{2,k}$.

For $\mathbf{C}_{3,k}$, according to the commutativity of matrix trace, it can be yielded that

$$\begin{aligned} \mathbb{E}\{\text{Tr}(\mathbf{C}_{3,k})\} &= w^{-1} \mathbb{E}\{f_{sc}^T f_{sc}\} \text{Tr}\{\bar{\mathbf{A}} \xi \xi^T \bar{\mathbf{A}}^T\} \\ &= (t_f/w) \mathbb{E}\{f_{sc}^2\} \bar{\mathbf{a}}_l^T \bar{\mathbf{a}}_l \end{aligned} \quad (20)$$

where $\bar{\mathbf{a}}_l$ denotes the l th column of $\bar{\mathbf{A}}$, and $t_f \leq w$ is the fault duration within the sliding window.

Then, by combining (18) with (19) and (20), it can be derived that the control limit should satisfy

$$\delta_{sc} \geq \frac{t_f}{w} \frac{\mathbb{E}\{f_{sc}^2\}}{\sum_{i=1}^{n-1} \sigma_{\lambda_i}^*} \max_{l \in [1, n]} \{\bar{\mathbf{a}}_l^T \bar{\mathbf{a}}_l\}. \quad (21)$$

Equation (21) characterizes the quantitative threshold correlation between the designed fault indicator and the original fault signature. Once the normalized differential voltage variances and the sum of eigenvalue SDs for all the transformed differential voltage channels are obtained under fault-free conditions, the control limit self-commissioning can be achieved according to the allowable fault signature f_{sc} in turn.

III. SSC FAULT DIAGNOSIS METHOD

Similar to previous data-driven SSC fault diagnosis methods, the proposed fault diagnosis method also comprises fault detection and localization. During the SSC fault detection, it is expected that the designed fault indicator with better performance will be able to detect more incipient faults. In other words, when the tolerable amplitude threshold of the fault signature is evaluated and determined, the control limit of the fault indicator is expected to be comprehensively self-commissioned with the guidance of interpretable fault detectability. This is crucial for the advancement of more adaptable and intelligent battery management functions, where the fault-localization-based fault isolation is a key derivative.

In the following sections, the control limit self-commissioning is carried out based on the quantitative threshold correlation between the designed fault indicator and the original fault signature. This forms the backbone of the SSC fault detection procedure, allowing for the evaluation of detectable SSC fault signatures concurrently. Then, the fault localization analysis is conducted based on the eigenvector contribution evaluation. Indeed, from the proposed improved virtual voltage measurement scheme in minimizing parameter-inconsistency-induced differential voltage fluctuations on windowed voltage streaming from both temporal and spatial dimensions, further efficient full-dimensional eigenvalue decomposition directly links the complementary fault detection and fault localization in the proposed SSC fault diagnosis method by contributing to the fault localization with accompanying eigenvectors, and

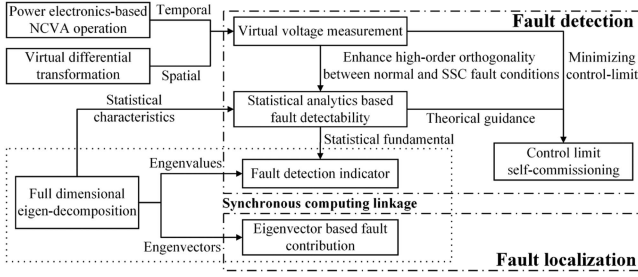


Fig. 3. Interdependency diagram of the proposed method.

their interdependency is outlined in Fig. 3. This synchronous computing linkage between fault detection indicator and fault contribution evaluation is beneficial for improving the SSC fault diagnosis performance.

A. Control Limit Self-Commissioning

Based on the SSC fault detectability of the designed indicator, as detailed in Section II-D, once the permissible SSC fault signature is determined prior, the fault detection control limit can then be evaluated as follows.

Considering that the cumulative SSC fault signature, characterized by differential voltage biasing, is influenced not only by the time-varying R_{sc} but also by the SOC variation during SSC fault occurrence, the fault signature f_{sc} occurring at the l th channel can be determined by

$$f_{sc}(t_2) \approx f_{ocv}(SOC) + \frac{V_l^{sv}(t_2)}{R_{sc}(t_2)} R_{s,l}(t_2) \quad (22)$$

where

$$f_{ocv}(SOC) = V_{oc,l}^{sv}(SOC(t_1)) - V_{oc,l}^{sv}(SOC(t_2)) \quad (23)$$

and t_1 represents the starting time of the SSC fault while the time interval $[t_1, t_2]$ denotes the SSC fault duration, and time-varying R_{sc} is the equivalent SSC resistance when momentary or cumulative SSC fault occurs.

Afterward, the final control limit δ_d can be determined by

$$\delta_d = \max\{\delta_{sc,1}, \delta_{sc,2}\} \quad (24)$$

where the theoretical control limit $\delta_{sc,i}$ ($i = 1, 2$) corresponding to the two virtual measurement coefficient submatrix $\mathbf{A}_{d,i} \mathbf{A}_v$ in the presence of fault signature f_{sc} occurs at the l th channel s_l is bounded by

$$\delta_{sc,i} \geq \frac{t_f}{w} \frac{\mathbb{E}\{f_{sc}^2\}}{\sum_{j=1}^{n-1} \sigma_{\lambda_j^*}} \max_{l \in [1, n]} \{\bar{\mathbf{a}}_{l,i}^T \bar{\mathbf{a}}_{l,i}\} \quad (25)$$

where $\bar{\mathbf{a}}_{l,i}$ denotes the l th column of $\bar{\mathbf{A}} = \Sigma_0^{-1} \mathbf{A}_{d,i} \mathbf{A}_v$, and t_f is the fault duration within the sliding window.

Combined with (4), to ensure consistent fault detectability for all differential voltage channels, the entries of $\bar{\mathbf{A}}$ should satisfy that

$$\max_{l \in [1, n]} \{\bar{\mathbf{a}}_{l,i}^T \bar{\mathbf{a}}_{l,i}\} = 2 \max_{l \in [1, n]} \{\sigma_{x_l^*}^{-2}\} \quad (26)$$

for at least one virtual measurement coefficient submatrix. It dominates the entry selection for subvectors $\Delta \mathbf{V}_{1,k}^{sv}$ and $\Delta \mathbf{V}_{2,k}^{sv}$. As an intuitive solution, the entry clusters Φ_1 and Φ_2 corresponding to the two subvectors can be simplified as

$$\begin{cases} \Phi_1 = \{1, 2, \dots, n-1\} \\ \Phi_2 = \{1, 2, \dots, \lfloor n/2 \rfloor, \lfloor n/2 \rfloor + 2, \dots, n\} \end{cases} \quad (27)$$

Note that, in either virtual measurement coefficient submatrix, there are always two columns, which have $\bar{\mathbf{a}}_{l,j}^T \bar{\mathbf{a}}_{l,j} = \sigma_{x_l^*}^{-2}$ ($j \in \{1, n\}$ for Φ_1 and $j \in \{\lfloor n/2 \rfloor + 1, \lfloor n/2 \rfloor + 2\}$ for Φ_2 in this case). The differential voltage channel variances corresponding to these two columns do not participate in the following control limit evaluation.

From (25) and (26), it is indicated that the control limit of the designed fault detection indicator is determined not only by the fault voltage amplitude f_{sc} but also by two important statistics: 1) the maximum normalized differential voltage variance, and 2) the sum of eigenvalue SDs for all transformed differential voltage channels under fault-free conditions. From the perspective of enhancing SSC fault detection performance, the improved virtual voltage measurement scheme plays a critical role by reducing the variances of normalized differential voltages, thereby facilitating the eigenvalue-decomposition-based fault diagnosis.

Note that, as the inverse operation of control limits self-commissioning, the control limit δ_d can be predetermined using a significance level α for the two streaming transformed w -width differential measurement matrices. This is possible through methods such as KDE [27] or simplified empirical method [34] under fault-free conditions, due to the designed indicator as (13) is free of Gaussian distribution. Then, the allowable SSC fault signature can be evaluated by using (22)–(27) in turn.

B. Fault Localization Analysis

Once the SSC fault is detected when the value of the designed detection indicator is beyond the self-commissioned control limit, the fault location can be further identified based on the eigenvalue contribution evaluation.

For the streaming differential measurement matrix $\bar{\mathbf{X}}_k$, the empirical eigenvalues $\{\lambda_{m,k}\}$ and corresponding eigenvectors $\{\mathbf{v}_{m,k}\}$ of its covariance matrix $\mathbf{C}_k$ satisfy

$$\mathbf{C}_k \mathbf{v}_{m,k} = \lambda_{m,k} \mathbf{v}_{m,k}, \quad m = 1, \dots, n-1 \quad (28)$$

where $\mathbf{v}_{m,k}$ is the m th column of the eigenvector matrix.

Inspired by the work in [35], the contribution of i th row of a covariance matrix to the eigenvalue $\lambda_{m,k}$ can be measured by the i th entry of the corresponding eigenvector $\mathbf{v}_{m,k}$. Then, the SSC fault localization can be characterized by normalizing the eigenvector contributions corresponding to outlier eigenvalues $\lambda_{j,k}$ when $\text{DI}_k = \max |\bar{\lambda}_{j,k}| > \delta_d$

$$\eta_i = (v_{j,k}^{(i)})^2, \quad i = 1, \dots, n-1 \quad (29)$$

where $v_{j,k}^{(i)}$ denotes the i th entry of $\mathbf{v}_{j,k}$ regarding $\lambda_{j,k}$. Note that, $\eta_i \in (0, 1)$ and $\sum_i \eta_i = 1$.

Afterward, the fault contributions to the streaming differential voltage vector as (7) under the SSC fault condition can be evaluated. Taking the entry clusters selected by (27) as an illustration,

the final contribution vector η_t can be summarized as

$$\eta_{i,t} = \begin{cases} 0.5(\eta_{i,\Phi_1} + \eta_{i,\Phi_2}), & i \in [1, \lfloor n/2 \rfloor] \\ 0.5\eta_{i,\Phi_1}, & i = \lfloor n/2 \rfloor + 1 \\ 0.5(\eta_{i,\Phi_1} + \eta_{i-1,\Phi_2}), & i \in [\lfloor n/2 \rfloor + 2, n) \\ 0.5\eta_{i-1,\Phi_2}, & i = n \end{cases} \quad (30)$$

For the SSC fault, if the two largest elements $\{\eta_{i,t}, \eta_{j,t}\}$ satisfy

$$\begin{cases} j = n, & i = 1 \\ j = i - 1, & i = 2, \dots, n \end{cases} \quad (31)$$

and

$$\begin{cases} \eta_{i,t} + \eta_{j,t} \geq \eta_{th} \\ \eta_{i,t} \geq \eta_{th}/4, \eta_{j,t} \geq \eta_{th}/4 \end{cases} \quad (32)$$

where η_{th} is the fault contribution threshold. Then, it can be confirmed that the fault location is the i th cell.

As analyzed in Section II-B, the differential voltage spike amplitude is still related to duty-cycle current magnitude, although the impact of parameter inconsistencies, as evidenced in Fig. 2, can be minimized by the improved virtual voltage measurement scheme. Consequently, significant “pseudo” fault signatures with duty-cycle-related differential voltage spikes are an expected occurrence, particularly under high-CR operating conditions. In this circumstance, false control-limit triggering is inevitable. Nevertheless, it is pragmatically posited that the IR diversity among individual LiB cells is relatively uniformly distributed. Then, the eigenvector contributions as (29) are likely to exhibit a discrete distribution characterized by quasi-synchronized changes in differential voltage variances. From this standpoint, the obtained SSC fault localization time-series can be considered as a subset of fault detection triggering time-series. As a result, the false alarm rate of SSC fault detection with (13) can be substantially decreased with the complementary fault localization process discussed in this section. More critically, it implies that the subdatasets containing only lower CR operating points can be extracted from historical datasets by lowering the permissible fault signature amplitude f_{sc} . Equivalently, this allows for a further reduction in the fault detection control limit for the proposed method, without adversely impacting its fault diagnosis efficacy.

C. Complete SSC Fault Diagnosis Procedure

Drawing from the comprehensive analysis presented in the earlier sections, Fig. 4 illustrates the complete SSC fault diagnosis procedure. This proposed methodology encompasses four primary stages: initiating with the improved virtual voltage measurement scheme, followed by control limit self-commissioning, and culminating with online fault detection and localization.

Initially, the virtual voltage measurement scheme requires the determination of aligned near-cut-off voltage to reduce unexpected differential voltage fluctuations due to parameter inconsistencies in the collected fault-free historical measurement voltage datasets. Subsequently, streaming transformed differential voltage matrices are generated by further virtual transformation. During the control limit self-commissioning stage, the reference parameters are extracted from the obtained

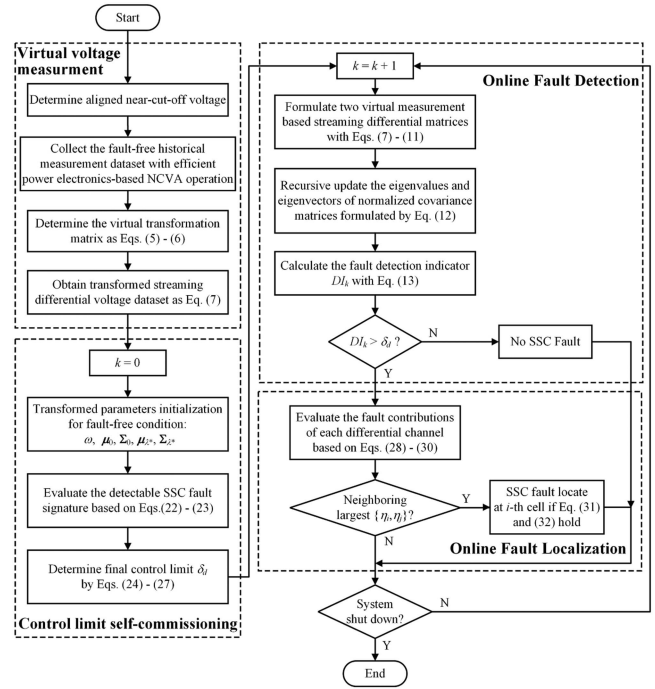


Fig. 4. Flowchart of the SSC fault diagnosis method.

TABLE I
SPECIFICATIONS OF ANR26650M1B CELL

Item	Value	Item	Value
Nominal voltage/V	3.3	Nominal capacity/Ah	2.5
IR/mΩ	6	Standard charge rate	1C
Voltage Range/V	[2.0,3.6]	Maximum discharge rate	28C

virtual transformed fault-free datasets as detailed in Section II-C. On this basis, the detectable fault signatures can be offline evaluated to characterize the fault detectability, leading to the finalization of control limit self-commissioning, as delineated in Section III-A. Following this, the designed indicator is computed in the online fault detection stage. Upon triggering the fault localization function, the fault contributions are evaluated as described in Section III-B to pinpoint the SSC fault location, and the miss-triggering can be effectively rectified.

IV. EXPERIMENT SETUP AND FAULT INJECTION

A. Experiment Platform

To investigate the characteristics of LiB cells under SSC fault conditions, the experiments were conducted on a compact LiB pack composed of 14 series-connected cells. This compact LiB pack can effectively mimic the operational dynamics of vehicular packs, with adjacent LiB cells interconnected via nickel strips through spot welding. The specifications of LiB cells employed in this study are detailed in Table I.

Fig. 5 depicts the hardware setup equipped with a commercial BMS prototype for experimental validation. This prototype incorporates an MC33771-based analog-front-end (AFE) and a switching control module comprising the

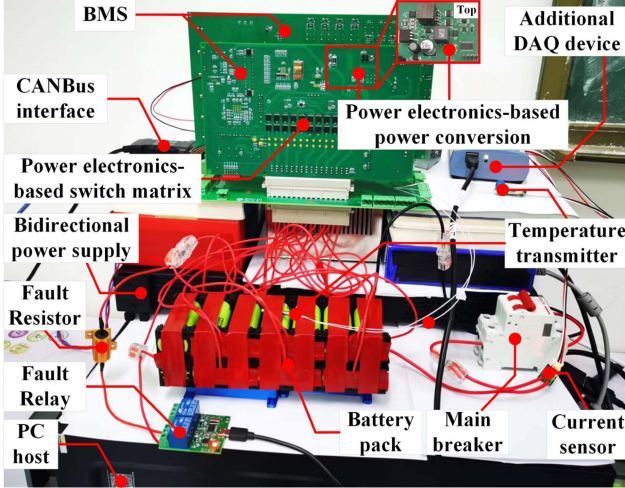


Fig. 5. Hardware setup for experimental validation.

“EMB1428+EMB1499” combination, designed for cell monitoring and active cell balancing (ACB) as recommended in Section II-B. This power-electronics-based ACB configuration benefits from switch-matrix driving in conjunction with isolated bidirectional power conversion to achieve efficient NCVA operation. The core of the BMS, a TMS320F28075 microcontroller, facilitates data collection and communication-based control with the AFE and ACB chipsets at a sampling frequency of 10 Hz.

Furthermore, data collected is transmitted to a PC host via the CANBus protocol, where the datasets are stored and analyzed. Moreover, the pack current and temperature data are concurrently sampled and transmitted using a USB-1608G data acquisition (DAQ) device, equipped with an ACS712 current sensor and a PT-100 thermocouple. An IT-M3422 bidirectional dc power supply integrating source and sink functions is utilized to charge and discharge the LiB pack according to the driving duty-cycle based on the World Light Vehicle Test Cycle (WLTC). With a standard charge rate of 1 C, the maximum discharge rate is limited to 2 C (5 A).

The momentary and cumulative fault injection simulation method, employed to replicate the time-varying SSC fault response, is elaborated upon in the subsequent section.

B. Momentary and Cumulative Fault Injection

Generally speaking, SC faults can be broadly categorized into internal SC (ISC) faults and external SC (ESC) faults. Regarding ESC faults, they are very pronounced in both fault characteristics and intensity, which can be considered as momentary hard faults and are easily detected by BMS at the module or pack level. Conversely, ISC faults are concealed within the complex dynamics of batteries and have the potential to deteriorate into a hard SC fault. Consequently, both momentary and cumulative soft faults are considered for ISC scenarios. As the triggering mechanism of ISC faults is still an open problem, a flexible data-driven approach is employed to simulate the time-varying

resistance effect in parallel across the target LiB cell, as opposed to utilizing a fixed resistor array.

Based on the typical static OCV and IR curves drawn in Fig. 2(a), the fault voltage of i th LiB cell at time index $k + t$ if SSC fault occurred at time index k can be formulated as

$$V_{i,k+t}^{SSC} = \frac{V_{i,k+t}^c - \Delta V_{oc,est,k+t}}{1 + R_{s,i,k+t}^{pd}/R_{sc,i,k+t}} \quad (33)$$

where $R_{sc,i,k+t}$ denotes the SSC equivalent resistance at time index $k + t$, $R_{s,i,k+t}^{pd}$ stands for the dynamic resistance considering polarization effect; $\Delta V_{oc,est,k+t}$ is the simulated OCV bias at time index $k + t$ due to SSC fault, which can be estimated as

$$\Delta V_{oc,est,k+t} = V_{oc}(C_{\Delta,k+t}^N) - V_{oc}(C_{\Delta,k+t}^{SSC}) \quad (34)$$

where $V_{oc}(C_{\Delta})$ characterizes the typical Ah-OCV mapping; the remaining Ah values $C_{\Delta,k+t}^N$ and $C_{\Delta,k+t}^{SSC}$ under the normal condition and the SSC fault condition, respectively, can be calculated as

$$\begin{cases} C_{\Delta,k+t}^N = C_{\Delta,k-1}^N + \sum_{j=1}^t I_{p,k+j} \cdot \Delta t \\ C_{\Delta,k+t}^{SSC} = C_{\Delta,k-1}^N + \sum_{j=1}^t (I_{p,k+j} + I_{ssc,k+j}) \cdot \Delta t \end{cases} \quad (35)$$

where $C_{\Delta,k-1}^N$ represents the remaining Ah value at the time index $k - 1$ just before the SSC fault occurred. $I_{ssc,k+j} = V_{i,k+t}^{SSC}/R_{sc,i,k+t}$ is the superimposed current due to SSC fault; Δt denotes the sampling period. Note that, (33)–(35) also hold for the normal condition when $R_{sc,i,k+t} \rightarrow \infty$.

Here, the momentary SSC fault injection is demonstrated by manipulating a fault relay series-connected to an equivalent fault resistor, as depicted in Fig. 5. The momentary SSC fault response upon the fault resistor switching is compared with the streaming data-driven modification based on (33) and (34) in Fig. 6 to validate the effectiveness of time-varying SSC fault simulation. Specifically, the momentary SSC fault is imposed on the No. 7 cell, with the fault response simulation relying on streaming voltages from Cell No. 6, owing to the superior cell consistency between these two cells. It is evident that the time-varying SSC fault simulation procedure is applicable for emulating both momentary and cumulative SSC fault injections.

As demonstrated in [36], variational cell current can result in time-varying polarization voltage. Therefore, different from long-time cell inconsistency, distinct fault signature lasting for tens of minutes can be induced by this polarization voltage dynamics with a pronounced transient process when momentary SSC faults occur. Indeed, the transient differential voltage drop due to about 0.13-C fault current step is significant in the case of this momentary SSC fault injection with a 10 Ω fault resistor. This momentary SSC fault corresponds to the demarcation between intermediate middle SC and early-stage SC, both of which are designated as SSC faults [37]. As illustrated in Fig. 6(a), the dynamic resistance, taking into account the polarization voltage dynamics, can be modeled using a modified second-order response as

$$R_{s,i,k+t}^{pd} = R_{s,i,k+t}^0 \cdot R_{s,i,k+t}^1 + R_{s,i,k+t}^2 \quad (36)$$

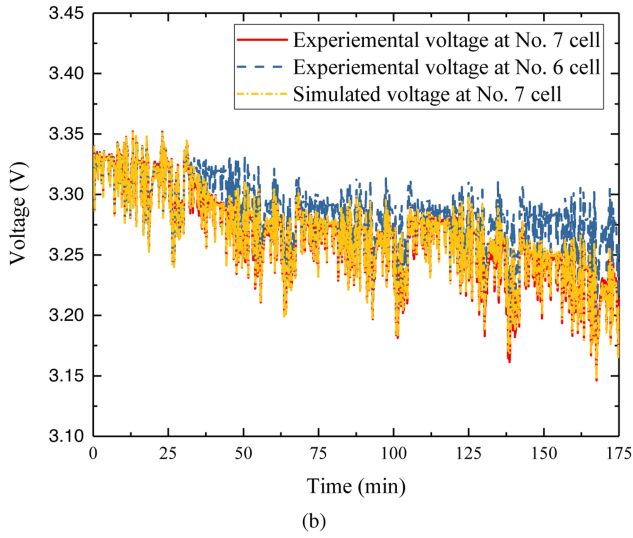
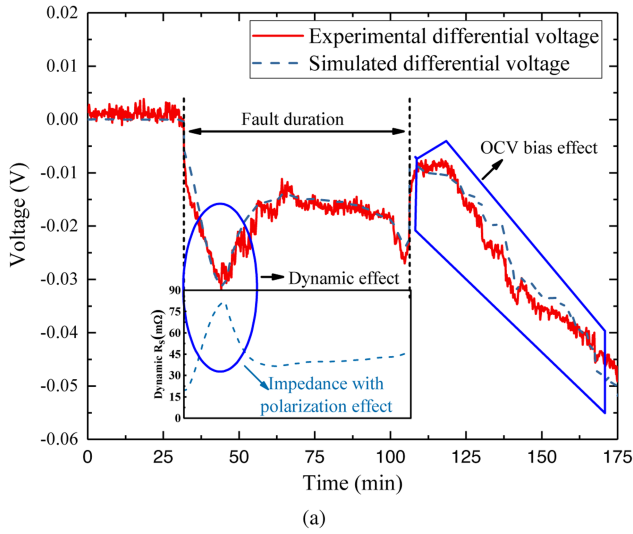


Fig. 6. Experimental and simulation comparison for momentary SSC fault injection with fixed fault resistor. (a) Differential voltage response curves. (b) Cell voltage response curves.

where

$$\begin{cases} R_{s,i,k+t}^0 = \frac{k_0}{\zeta_0} \cdot (1 - e^{-\zeta_0 \omega_n t}) \cdot \sin(\zeta_0 \omega_n t + \arctan \frac{\zeta_0}{\zeta}) \\ R_{s,i,k+t}^1 = k_2 \cdot e^{-k_3 t}, \quad R_{s,i,k+t}^2 = k_4 t \end{cases} \quad (37)$$

where $R_{\text{flt,step}}$ is the steady-state polarization resistance of battery cell, $k_0 = k_1 \cdot R_{\text{flt,step}}$ and $k_4 = k_5 \cdot R_{\text{flt,step}}$; $\zeta_0 = \sqrt{1 - \zeta^2}$. Although the polarization impedance is nonlinear and time-varying, the simulation residual with constant steady-state polarization resistance is relatively small and accepted, as compared in Fig. 6(a). In our LiB pack configuration, we have $k_1 = 5.67$, $k_2 = 8$, $k_3 = 0.0026$, $k_5 = 1/2400$, $\zeta = 0.4$, and $\omega_n = 0.0038$.

Indeed, the steady-state equivalent impedance following an SSC fault occurrence is significantly larger than the inherent IR under SSC fault conditions. It relies on the fault configuration and pack topology, which can be divided into five counterparts,

TABLE II
COUNTERPARTS OF SC-FAULT-RELATED IMPEDANCE

Item	Value(mΩ)	How to obtain
Static IR	~ 6	YR1035 Tester
Nickel-strip connection resistance	~ 6.6	YR1035 Tester
Filtered dynamic IR	[8, 17]	Fig. 2a
Fault connection resistance	~ 7	YR1035 Tester
Steady-state polarization resistance	~ 8	Datasheet and [38]

as summarized in Table II for this specific fault injection scenario. Therefore, the detectable SSC fault voltage is primarily influenced by the steady-state equivalent impedance other than static IR.

Further examination of Fig. 6(a) reveals that the differential voltage is dominated by OCV bias after the fault duration. Thus, the cumulative effect resulting from an SSC fault cannot be immediately eliminated even with an optimally configured cell-balancing circuit in place. As a result, the abnormal cell inconsistency induced by the SSC fault should be continuously alarmed by the fault diagnosis logic.

V. RESULT AND DISCUSSION

In this section, we initially validate the improved virtual voltage measurement scheme, which incorporates efficient power-electronics-based NCVA operation, by contrasting it with the conventional standard voltage measurement scheme. Building on this foundation, the SSC fault diagnosis performance of the proposed method is then extensively evaluated against two contemporary state-of-the-art data-driven methods: 1) the voltage-cosine-similarity-based method (VCS) in [22] and the voltage-correlation-based independent component analysis (ICA-VC) method in [27]. Subsequently, we delve into the impact of reducing the fault detection control limit on the overall fault diagnosis performance.

A. Comparison of Voltage Measurement Schemes

In this section, three cases with different voltage measurement schemes are compared to illustrate the contribution of an improved voltage measurement scheme to minimize parameter-inconsistency-induced differential voltage fluctuations. In case 1, by employing the WLTC duty-cycle with a maximum cycle current of 2 C after standardized charging to 100% SOC, the differential voltage curves with original channel order for multiple cycles without cell balancing are collected and presented in Fig. 7(a). In case 2, the configured power electronics-based ACB function is activated when the pack voltages approach $2.94 \times 14 = 41.16\text{V}$ for NCVA operation, and the LiB stack is then recharged to 100% SOC without further cell balancing. Here, this process is denoted as NCVA charging. Afterward, by applying the same duty-cycle as case 1, the differential voltage curves still with the original channel order for multiple cycles without cell balancing are illustrated in Fig. 7(b). On this basis, the nonzero elements in the virtual transformation matrix $\mathbf{A}_v$ in (7) can be determined based on binary-encoding-based discrete swarm intelligence optimization. Accordingly, the optimized channel order in case 3 can be obtained as [10, 2, 1, 14, 12,

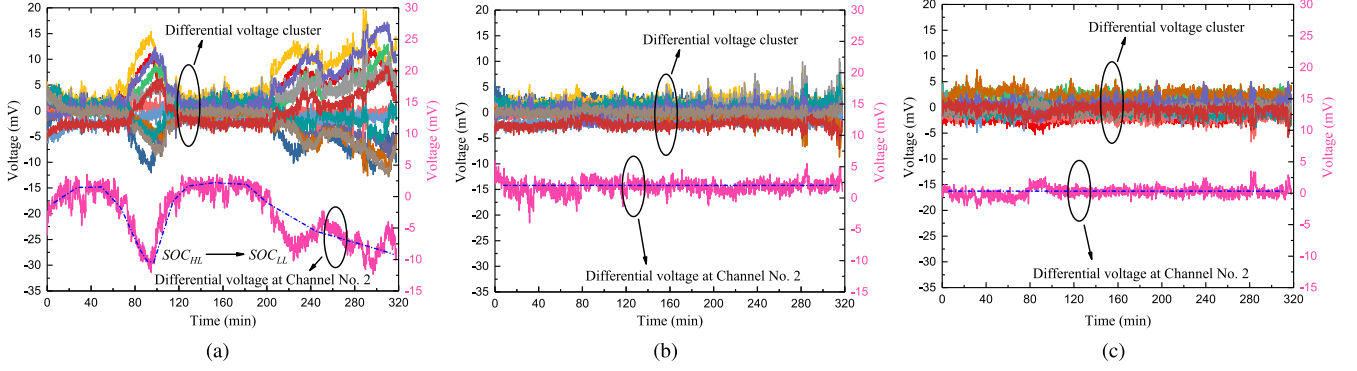


Fig. 7. Comparison of different voltage measurement schemes. (a) Original voltage measurement after standardized charging. (b) Original voltage measurement after NCVA charging. (c) Virtual voltage measurement with NCVA charging.

TABLE III
SDs OF DIFFERENTIAL CHANNELS

No.	SD (mV)			No.	SD (mV)		
	Case 1	Case 2	Case 3		Case 1	Case 2	Case 3
1	3.380	0.742	0.923	8	1.088	0.732	1.014
2	4.073	1.030	0.742	9	2.911	1.208	0.995
3	4.284	1.152	0.837	10	2.314	0.927	0.893
4	3.269	0.990	0.816	11	4.761	1.029	1.122
5	2.318	0.997	0.811	12	3.367	0.634	0.814
6	2.181	1.435	0.999	13	2.975	0.749	0.997
7	0.932	1.078	0.828	MD	0.880	0.170	0.094

7, 4, 9, 3, 8, 11, 13, 5, 6], and the resulting differential voltage curves are showcased in Fig. 7(c). Here, the SOC range for all three cases is identical as [9.66%, 93.01%]

As compared in Fig. 7, considerable differential voltage fluctuations can be observed for case 1. The differential voltage curve for channel No.2, adjusted by a -20 mV bias, exemplifies this, displaying a trend consistent with that seen in Fig. 2(b). This concurrence validates the mechanism analysis conducted in Section II-A.

Furthermore, the SDs and mean deviation (MD) for all differential voltage channels under the three cases are summarized in Table III, highlighting the maximum and minimum SDs for each case and the minimal MD across all cases in bold. Evidently, the SDs in case 1 are significantly higher than those in the subsequent cases. Notably, the channel MD in case 1 is fivefold that in case 2, underscoring the importance of NCVA charging. Nonetheless, compared to case 2, the MD in case 3 from historical data, which covers approximately 83.3% of the full SOC range, is reduced by 45%, and the maximum SD has decreased by 22%.

Thus, in case 3, both the maximum SD and MD, which indicate channel voltage inconsistency, are concurrently minimized. Combined with (25) and (26), it can be ensured that the highest consistency in differential channel voltages significantly enhances fault detectability and optimizes sensitivity.

Normally, the duty-cycle-related differential voltage fluctuations induced by the cyclic current are not obvious under medium CR operating conditions (not more than 2 C in this

work), as shown in Fig. 7(b) and (c). The differential voltage fluctuation amplitudes are dominated mechanistically by the inherent parameter inconsistencies such as dynamic IR and capacity among individual cells. Then, the improved virtual voltage measurement scheme can effectively suppress these fluctuations through virtual cell pairing and approximately aligning the voltage plateau ranges, which may be unaligned resulting from inherent parameter inconsistencies. However, it is noteworthy that, more significant duty-cycle-related components can be imposed on the differential voltage fluctuations under high-CR operating conditions. In these circumstances, the permissible detectable fault signature f_{sc} needs to be higher according to (22) and (25). Nevertheless, f_{sc} can remain unchanged, and the fault diagnosis efficacy can be maintained with the complementary fault localization procedure, as discussed at the end of Section III-B.

As analyzed earlier, it can be confirmed from Fig. 7 that, the virtual transformation with constant matrix A_v is applicable in hour and even day time-scales. Nevertheless, it can be periodically updated on a larger time scale with incremental fault-free datasets based on the proposed deterministic SSC fault diagnosis method.

B. SSC Fault Diagnosis Performance

In this section, we present two scenarios to assess the SSC fault diagnosis performance of three comparison methods under both momentary and cumulative SSC fault conditions. Here, the validated fault injection simulation procedure is applied on faulty cell No. 11, with the assumption of constant steady-state polarization impedance, as outlined in Section IV-B.

By applying the inverse process of control-limit self-commissioning with (24) and (25), the detectable SSC fault voltage signature can be evaluated as $f_{sc} = 10.7$ mV with $\delta_d = 15.0414$ under the same confidence limit ($\alpha = 1 - 10^{-6}$) as in [27]. According to Table II, the steady-state $R_s \approx 40$ m Ω , and the voltage at the end of the improved virtual voltage measurement procedure in Fig. 7(c) when SOC $\approx 9.4\%$ is determined as $V^{sv} = 3.1$ V. Then, the minimum SC resistance is determined as $R_{sc,min} = 11.8 \Omega$ by (22) assuming at the beginning of fault injection with $f_{ocv}(\cdot) = 0$.

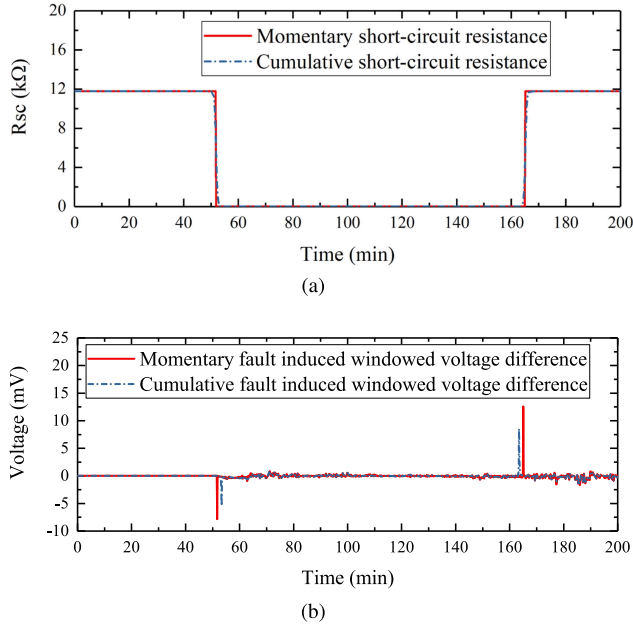


Fig. 8. Comparison of momentary and cumulative fault injections. (a) SC resistance curves. (b) Differential voltage response curves.

Note that, the momentary SSC fault can be regarded as a specific instant of cumulative SSC fault with a negligible cumulative period. For the cumulative SSC fault scenario, the battery aging effect can be disregarded, and the smooth SC resistance variation is analogously considered as a logistic process

$$R_{sc}(t) = \frac{K}{1 + \frac{K - N_0}{N_0} e^{(-rt)}} \quad (38)$$

where $K = R_{sc,max} - R_{sc,min}$ with $R_{sc,max} = 1000 \cdot R_{sc,min}$, $N_0 = K/2$, and $r = 0.0059$.

In both scenarios, a variable resistor R_{sc} is parallel connected with the faulty cell to simulate the two types of SSC fault injections. Under the normal condition, $R_{sc} = R_{sc,max}$. For the momentary SSC fault, the resistance instantly changes to $R_{sc} = R_{sc,min}$, whereas the transient logistic process as (38) is applied for the cumulative SSC fault. As demonstrated in Fig. 8(a), the fault injection in both scenarios is almost identical, exhibiting only minor transient variations. It is evident that the voltage difference becomes less pronounced with slower SSC fault progression, which is consistent with the gradual development of the SSC fault in its incipient stage [38]. However, the cumulative SSC fault scenario utilized in this study exhibits a similar level of windowed fault voltage difference as the momentary fault scenario, as shown in Fig. 8(b), using the same sliding window $w = 100$ for the proposed method and the ICA-VC method.

Subsequently, the SSC fault detection performance of the three methodologies is evaluated, with quantifiable diagnosis outcomes and key performance metrics summarized in Table IV. Here, DT represents the detection time, LA indicates whether there is a last alarm during fault injection, EW denotes warning against continuous fault signature regardless of fault removal, and FDR stands for the fault detection rate. $L = 68\,000$ s in

TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT METHODS

Fault case	Method	DT/s	Fault diagnosis performance			
			LA	EW	Cell No.	FDR/%
Momentary	ICA-VC	9.9	No	Yes	11	$20w/L$
	VSC	80.1	No	No	11	$10/L$
	Proposed	109.6	Yes	Yes	11	98.4
Cumulative	ICA-VC	649.8	No	No	-	0
	VSC	-	No	No	-	0
	Proposed	171.3	Yes	Yes	11	97.6

this study, which is the fault duration in seconds. It is important to note that, the proposed method leverages the orthogonality between the SSC fault component and the normal component, following a full-dimensional data rigid transformation based on the improved virtual measurement scheme. Consequently, robust sensitivity to both momentary and cumulative SSC faults at the millivolt-level fault signature can be guaranteed. However, quite diverse responses to momentary and cumulative SSC faults are exhibited by the recently proposed ICA-VC and VCS methods, although their momentary SSC fault detection time is less than the proposed method. Specifically, the FDRs in the momentary SSC fault scenario are only 0.0294% and 0.00015%, respectively, while the two methods do not respond to the injected cumulative SSC fault. This discrepancy primarily arises because the weak SSC fault signature with sufficiently slow progression cannot be effectively distinguished within the slighting window in the original measurement space. Moreover, the inherent cumulative effect of SSC fault signature is not fully exploited by the compared methods. There, in conjunction with the visualization comparisons in Fig. 9, these two methods lack sensitivity to millivolt-level SSC fault signatures.

Correspondingly, the visualization of fault localization using the proposed method is presented in Fig. 10 with $\eta_{th} = 0.8$. The fault contributions of each virtual measurement channel are displayed in Fig. 10(a), and the final fault diagnosis result pinpointing the original cell location is illustrated in Fig. 10(b). It is confirmed that both the momentary and cumulative SSC faults can be promptly detected and accurately localized, achieving high FDRs of 98.4% and 97.6%, respectively. As discussed at the end of Section IV-B, the signature of an SSC fault, attributable to its cumulative effect, persists even after the fault has been eliminated. This lingering effect can progressively diminish the performance of the faulty cell and, consequently, the entire LiB pack. Hence, continuous monitoring to ward off the cumulative impact of SSC faults is essential, as demonstrated in Fig. 10(b). However, the last-alarming SSC faults cannot be achieved by using ICA-VC and VSC methods. They inevitably misdiagnose the duration of fault signature due to its similar statistical characteristics such as high-voltage correlation and similarity to the normal condition.

It is noteworthy that, the polarization voltage dynamics resulting from external SSC resistance switching can enlarge SSC fault voltage signatures. Although the momentary SSC fault injection method with an external parallel-connected fault resistor is widely utilized, its applicability to the characterization

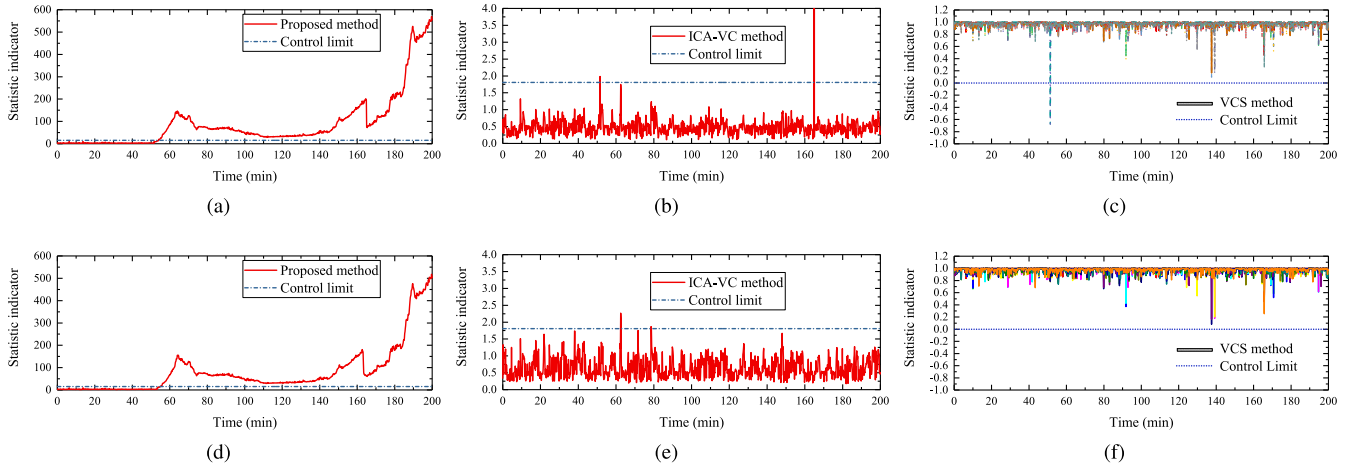


Fig. 9. Comparison of SSC fault detection performance. (a)–(c) Momentary fault conditions. (d)–(f) Cumulative fault conditions.

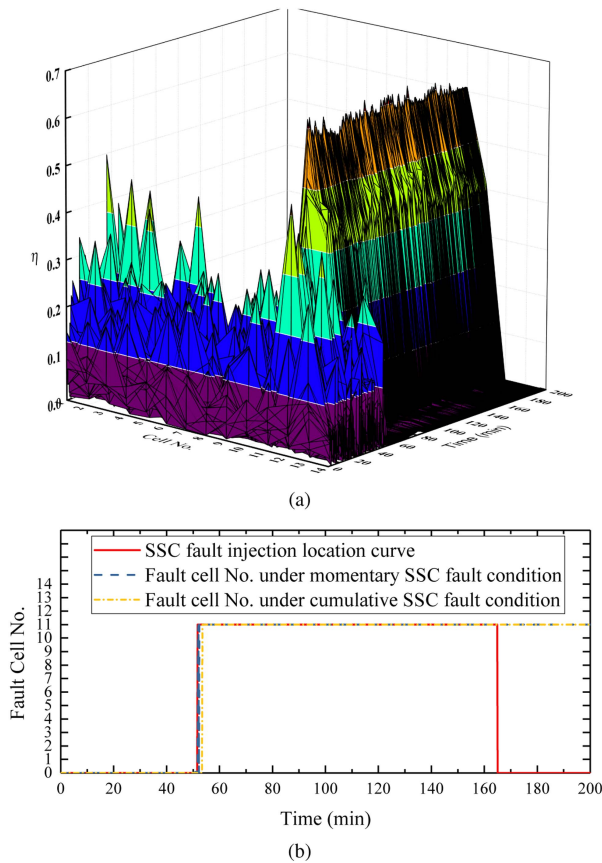


Fig. 10. Fault localization of the proposed method. (a) Fault contribution matrix. (b) Fault localization result.

of internal SSC faults still needs further investigation, which is beyond the scope of this article.

C. Influence of Fault Detection Control-Limit Reduction

To validate that fault alarm miss-triggering can be mitigated by complementary fault localization such that the fault detection

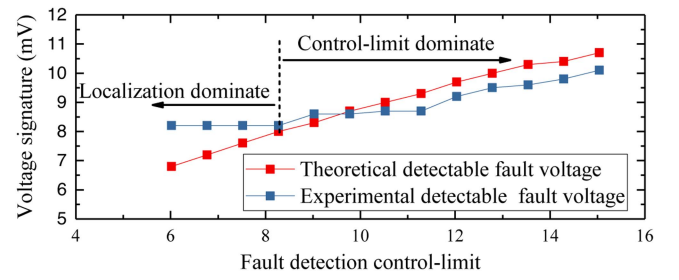


Fig. 11. Influence of fault detection control-limit regulation.

control limit can be reduced, the impact of reducing the fault detection control limit is evaluated herein.

As illustrated in Fig. 11, the fault localization correction plays a pivotal role in the fault diagnosis outcome when $\delta_d \leq 8.122$. Indeed, it is suggested that a lower confidence limit than the value obtained at the beginning of Section V-B for determining the fault detection control limit from historical datasets can be robustly configured. From this viewpoint, the effectiveness of the proposed method does not depend excessively on the quality of historical data.

By adjusting the confidence limit as $\alpha = 0.998$, 101.8 s ahead of fault detection without fault alarm miss-triggering can be achieved for the momentary SSC fault scenario while 106 s ahead fault detection for the cumulative SSC fault scenario. This represents a significant enhancement in fault diagnosis performance, with almost 61.9% advancement in SSC fault detection time compared to the results in Table IV.

It is important to note that the discrepancy between theoretical and experimental detectable voltage signatures, as observed in Fig. 11, primarily stems from substantial windowed differential voltage fluctuations due to high duty-cycle currents. Under such high-current conditions, the SDs of channel differential voltages are considerably larger than the values listed in Table III. These superimposed “pseudofault signatures,” as discussed in Section III-B, lead to not the single-channel fault distribution

initially assumed for SSC fault condition, but multichannel fault distributions as characterized by the fault contributions for the time interval [0, 53 min] in Fig. 10(a) with no SSC fault injection. Nevertheless, the control-limit-dominated difference between theoretical and experimental detectable fault voltage signatures is still no more than 6.5%. Therefore, it has no negative influence on the fault diagnosis performance of the proposed method.

VI. CONCLUSION

In this article, an interpretable full-dimensional statistical analytic method is proposed for SSC fault diagnosis of vehicular LiB battery packs. Minimized MD of SDs across differential voltage channels can be achieved by the improved virtual voltage measurement scheme with efficient power-electronics-based NCVA operation. By leveraging full-dimensional eigenvalue decomposition, the fault detection indicator is crafted, taking into account the high-order orthogonality between normal operation and SSC fault conditions. This method not only provides a theoretical foundation for fault detectability but also facilitates control limit self-commissioning. Furthermore, fault localization can be achieved with complementary eigenvector contribution evaluation to ensure the universal fault diagnosis robustness against both the momentary and cumulative SSC fault scenarios. By using the investigated SSC fault injection method, superior fault diagnosis performance than two state-of-the-art data-driven methods is achieved by the proposed method with the comprehensive consideration of polarization voltage dynamics and the cumulative effect of SSC fault, thereby enabling precise minute-level SSC fault diagnosis for both momentary and cumulative SSC fault conditions.

Nevertheless, more significant duty-cycle-related differential voltage fluctuations resulting from higher CR operation such as more than 4 C are inevitable for modern high-performance EVs, attributed to inherent parameter inconsistencies. In this circumstance, extracting a single-channel SSC fault signature is challenging, and the SSC fault diagnosis performance is difficult to be maintained. Future work will explore more effective methods for suppressing duty-cycle correlation, aiming for reliable SSC fault diagnosis under full duty-cycle conditions with the designed fault detection statistics and associated fault detectability theory.

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